



TARUMT
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UNIVERSITY OF
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VISION-BASED RIPENESS GRADING OF BANANA USING YOLO OBJECT CLASSIFICATION

BACHELOR OF MECHATRONICS ENGINEERING
WITH HONOURS

FINAL YEAR PROJECT 2

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Main Goal



To improve the speed and reliability of banana ripeness assessment in supermarket environments by integrating a YOLO-based classification system with CLAHE image preprocessing under unstable lighting conditions.

Objectives:



Develop a YOLO-based system for banana ripeness classification into four classes.



Apply preprocessing techniques to improve image quality under different lighting conditions.



Evaluate and compare the performance of the proposed methods.

Problem Statement

How can banana ripeness be detected reliably in practical conditions?

Background interference and lighting variation may reduce the reliability of banana ripeness classification.

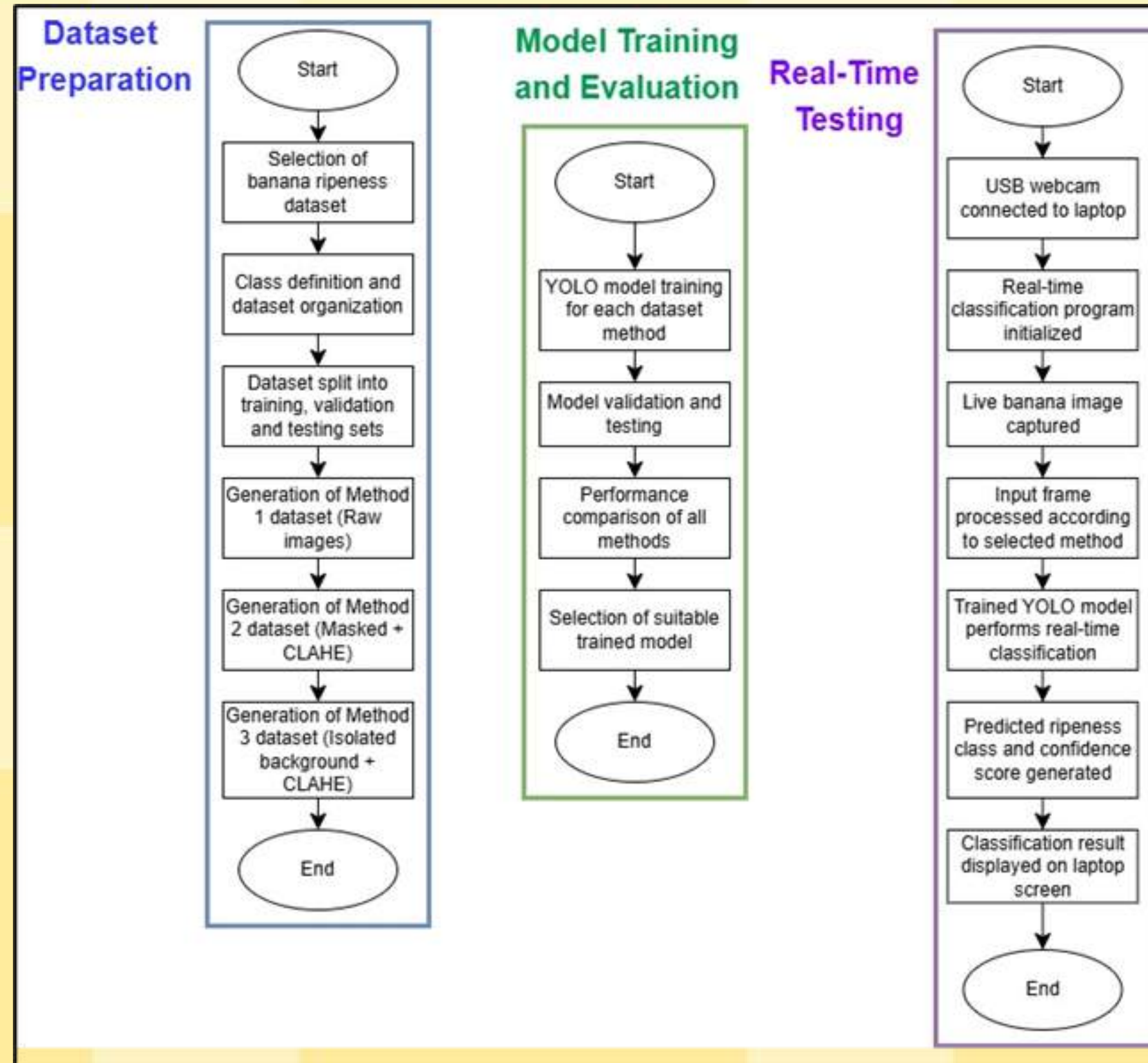


Lighting
Variation



Proposed Methodology

Three stage workflow with fair method comparison



Core experiment

➔ Common Experimental Setting

- Same source dataset for all methods
- Same class labels and dataset split
- Same YOLOv8 classification model

➔ Methods compared

- Method 1 Raw Images
- Method 2 Masked + CLAHE
- Method 3 Isolated background + CLAHE

➔ Evaluation

- Offline testing
- Real-time webcam testing
- Warm and cool lighting conditions

Dataset preparation methods

Visual differences between the three inputs



Method 1 Raw



Original images with natural background and lighting variation.



Method 2 Masked + CLAHE



Banana region emphasized with moderate background suppression.

Method 3 Isolated + CLAHE

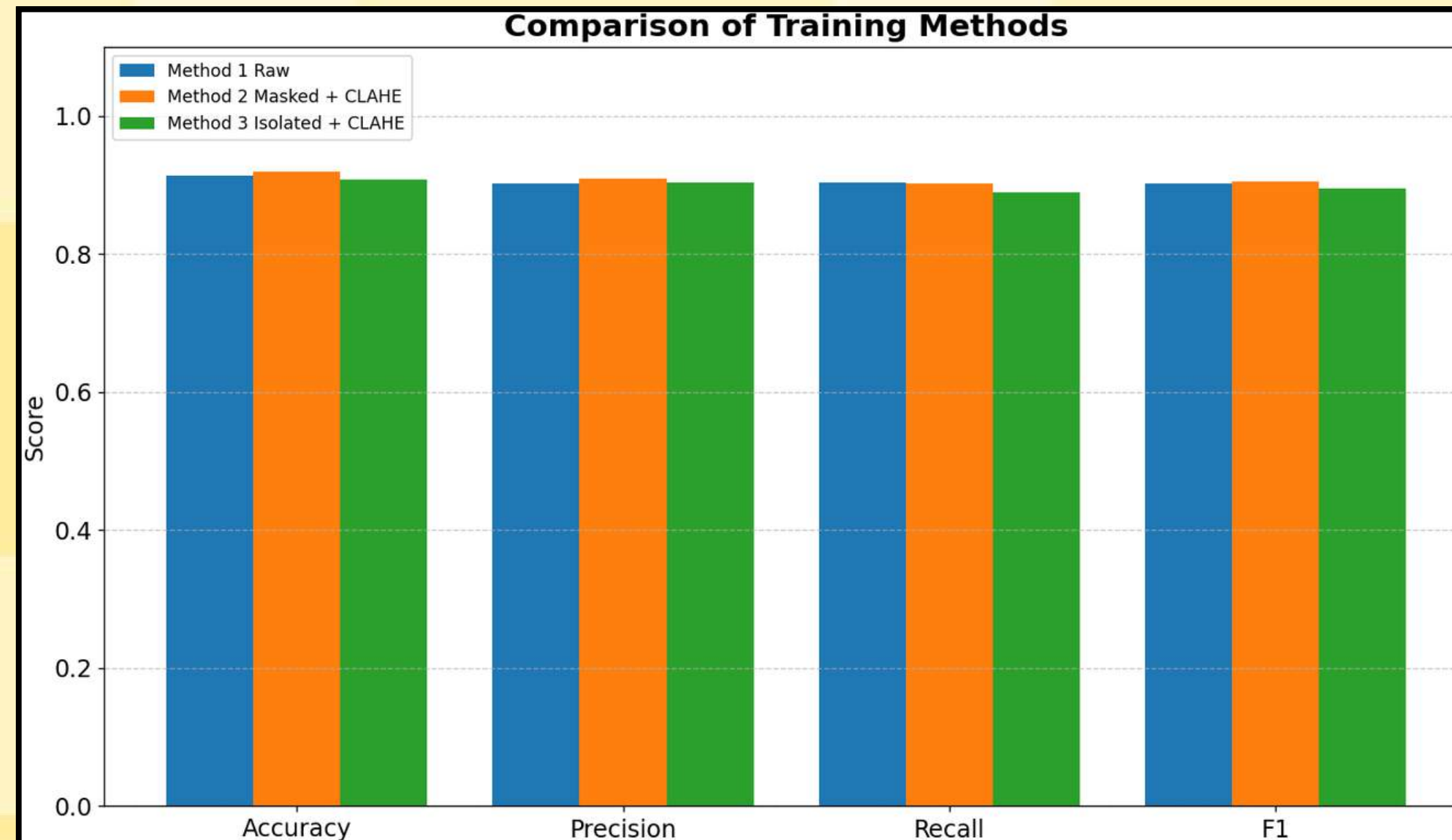


Background isolated to focus the model on the banana object.



Offline classification results

Method 2 performs best on the prepared test dataset



➔ High offline Accuracy

All three methods achieved more than 90% accuracy under offline testing

➔ Small Performance Gap

The difference among the three methods was relatively small in the prepared dataset setting.

➔ Best Offline Method

Method 2 (Masked + CLAHE) achieved the highest overall offline metrics.

➔ Strong Baseline Performance

Method 1 (Raw) still performed well, showing that the original images contained useful ripeness features.

➔ Limitation of Full Isolation

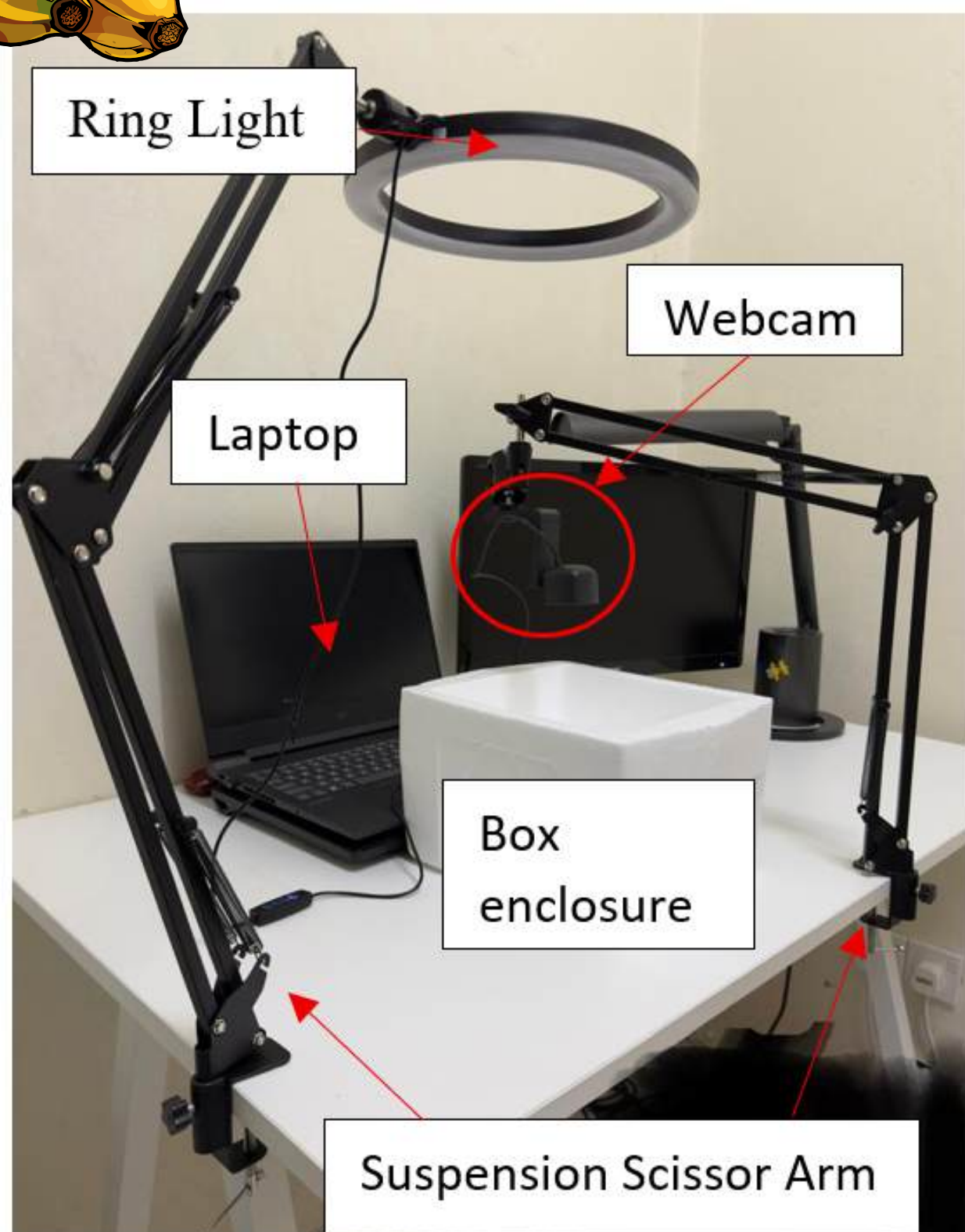
Method 3 was slightly lower offline, suggesting that full isolation may remove some contextual cues.



Design And Methodologies



Experimental Setup

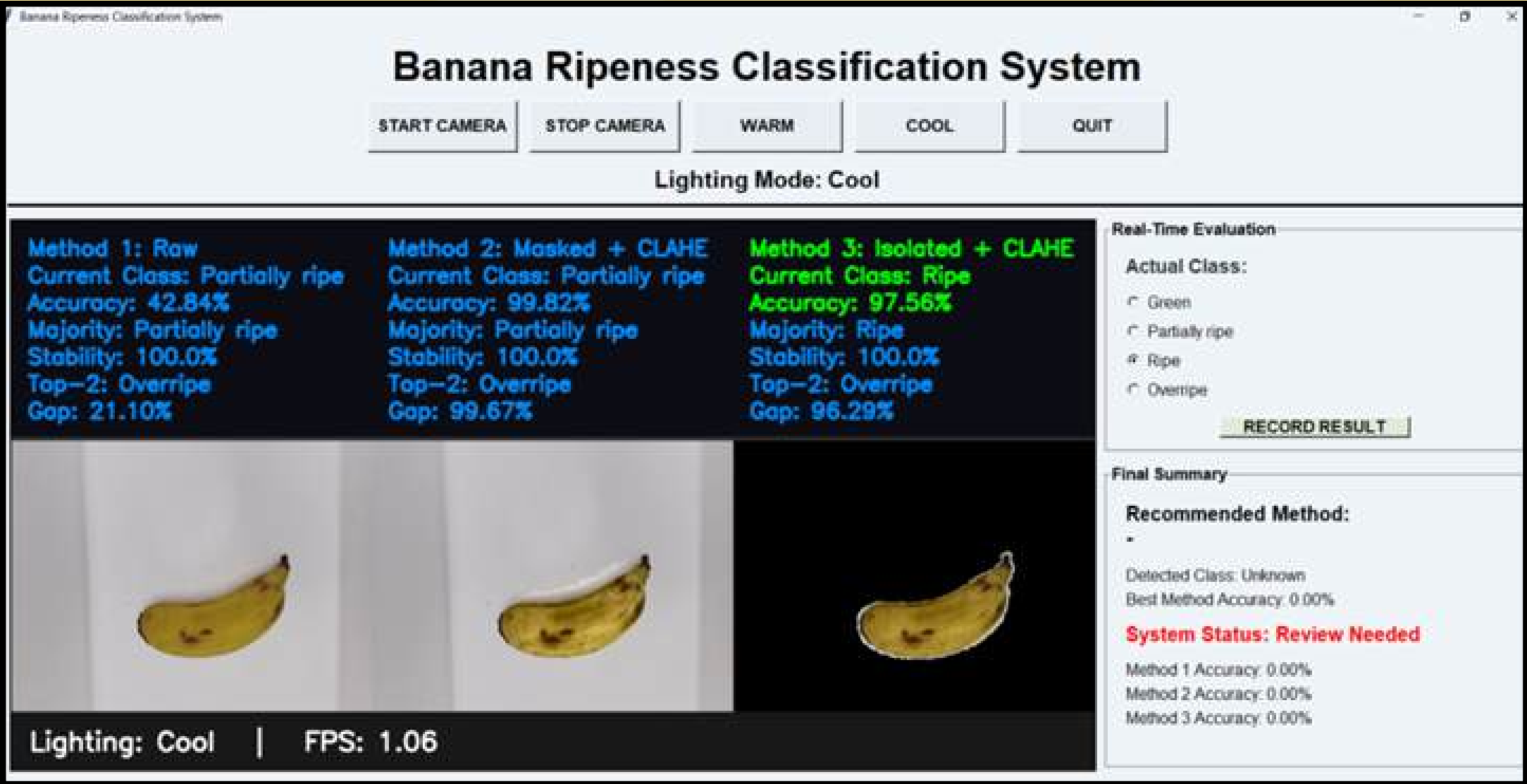
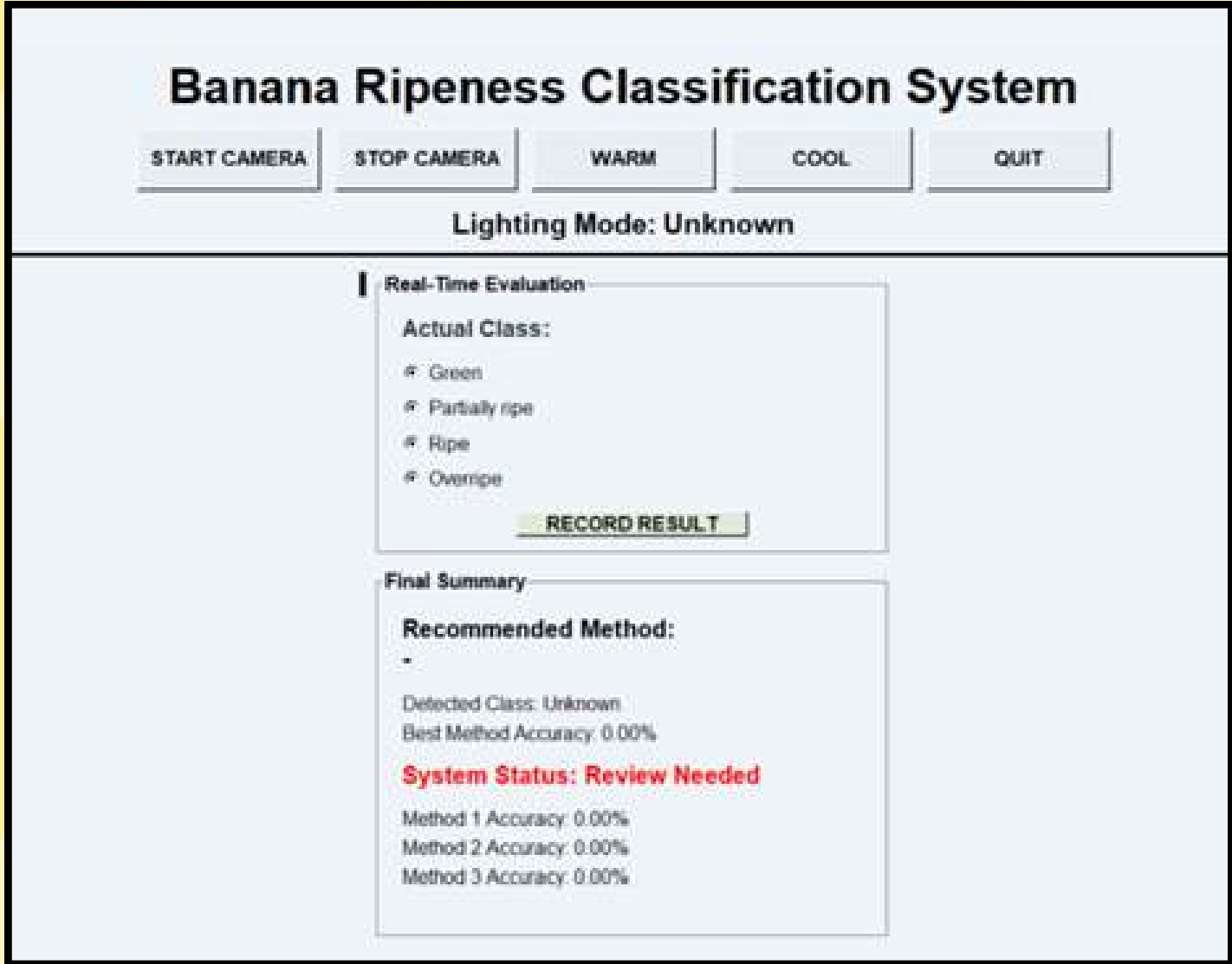


- **Webcam connected via USB to laptop**
- **Ring light used for warm and cool lighting**
- **Banana placed inside box enclosure for controlled image capture.**



Real-time testing platform

Custom GUI for live comparison under warm and cool lighting



GUI Functions

➔ Live Comparison

Displays the outputs of the three trained methods side by side in real time.

➔ Real-Time Evaluation

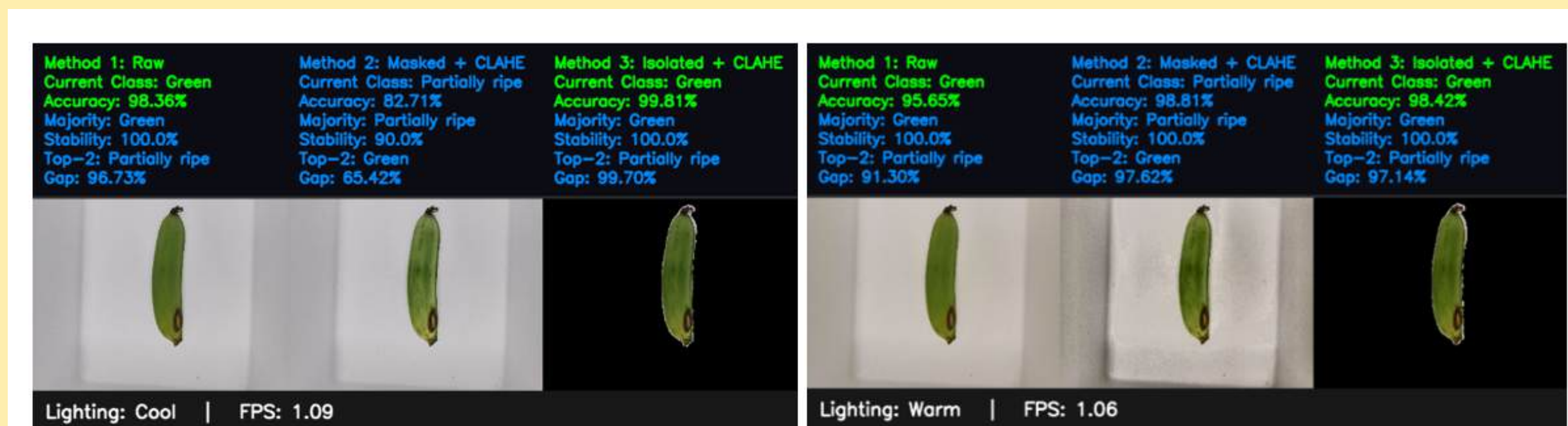
Shows predicted class and confidence, and highlights correct results above 80% in green.

➔ Automatic Recording

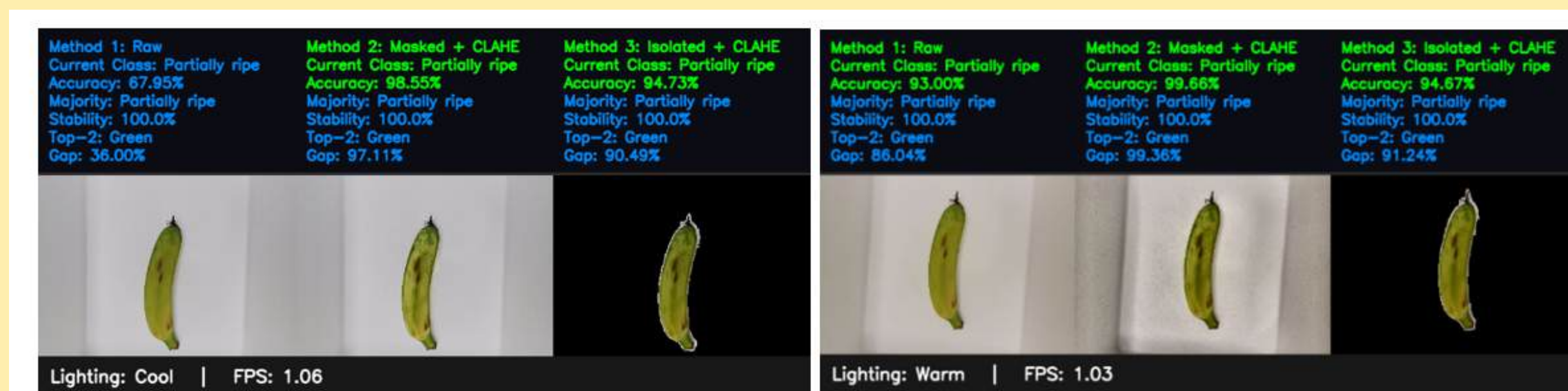
Saves screenshots and CSV results for later comparison and analysis.



Results And Discussion



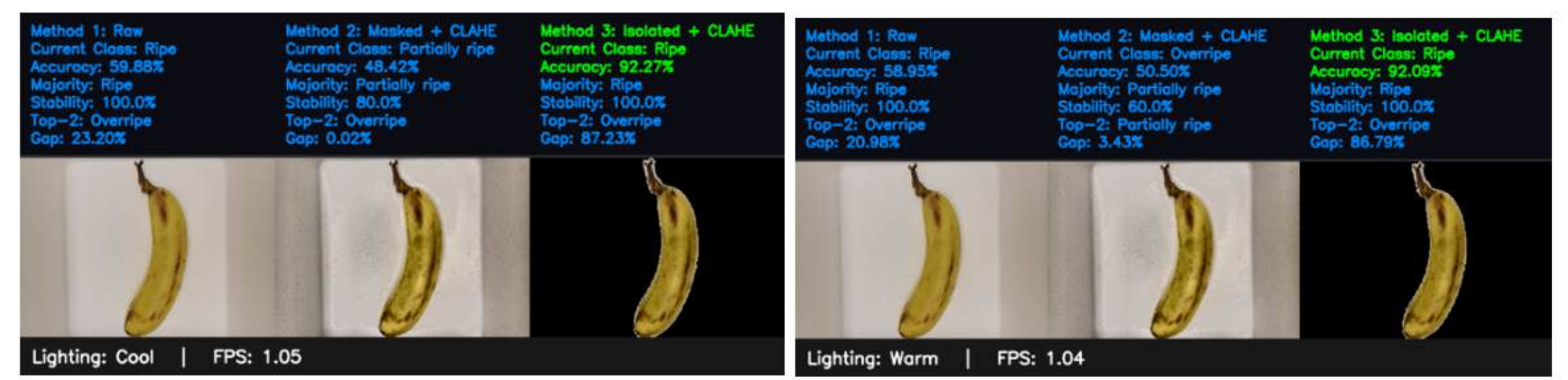
(a) Green Banana under cool and warm lighting



(b) Partially ripe banana under cool and warm lighting



Results And Discussion



(c) Ripe banana under cool and warm Lighting

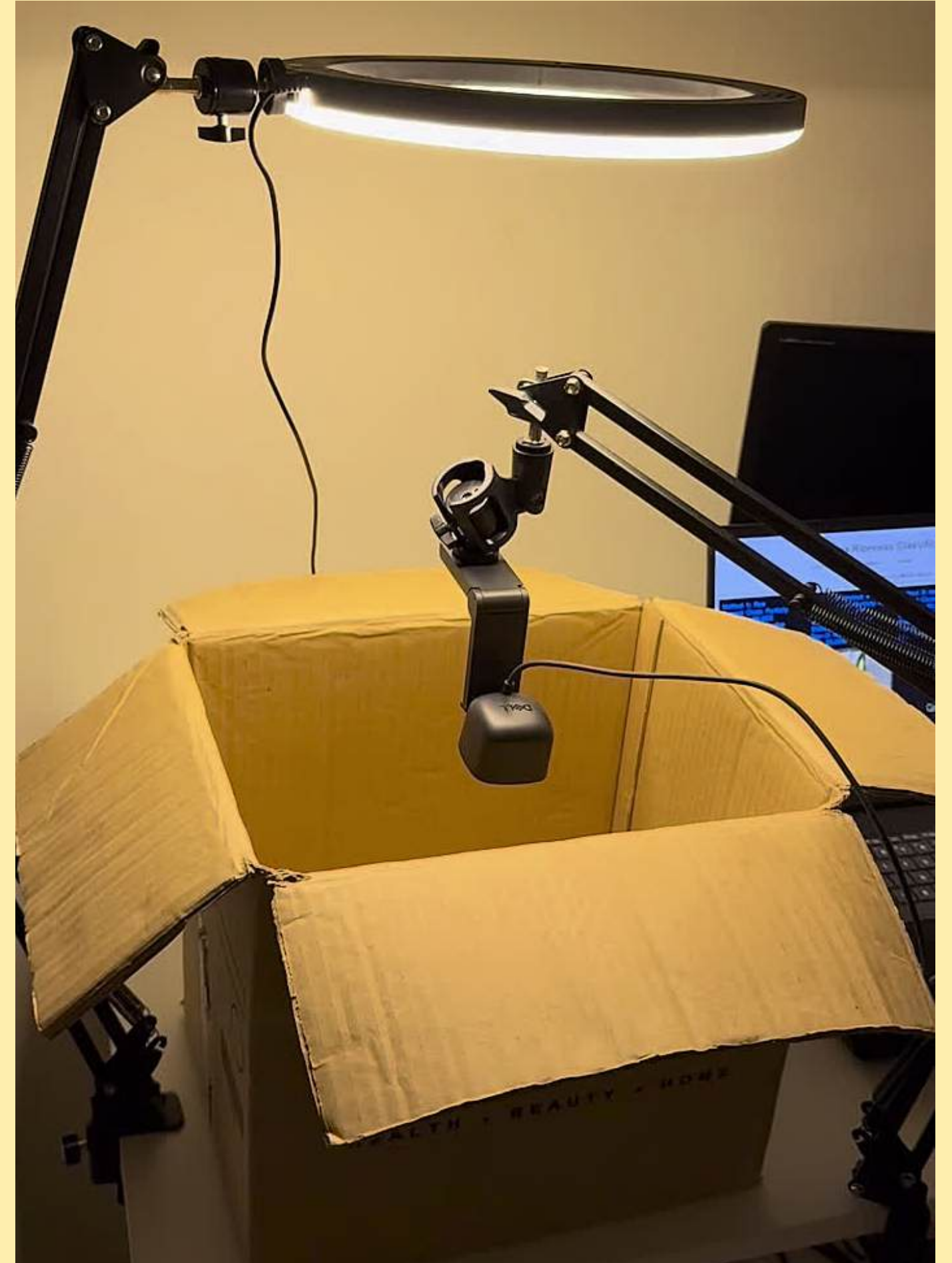


(d) Overripe banana under cool and warm Lighting

Unripe Banana Video



Partially-Ripe Banana Video



Ripe Banana Video

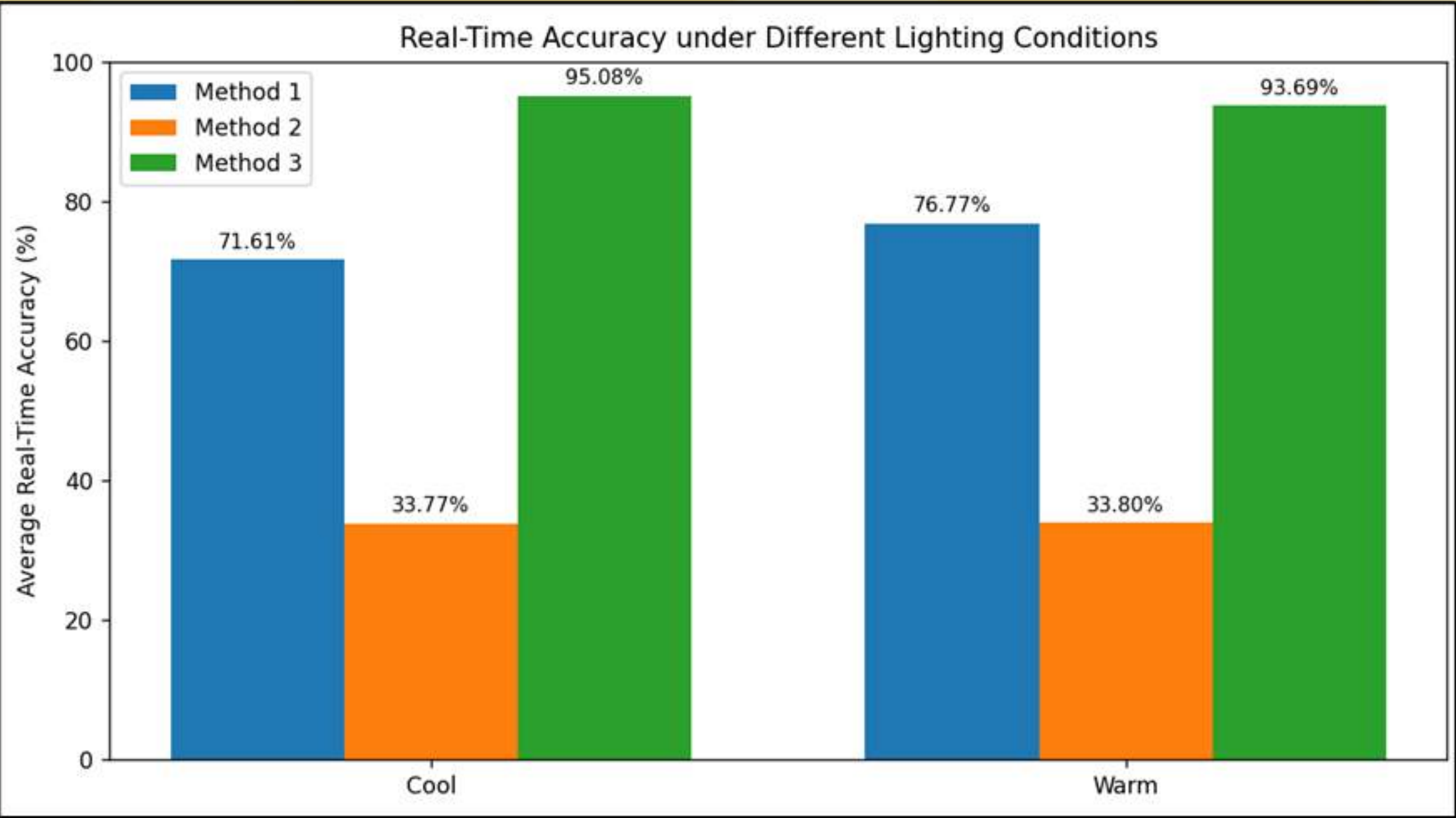


Overripe Banana Video



Real-time results under lighting variation

Method 3 becomes the practical winner



➔ Best real-time stability

Method 3 remained consistently strong under both lighting conditions.



➔ Moderate baseline performance

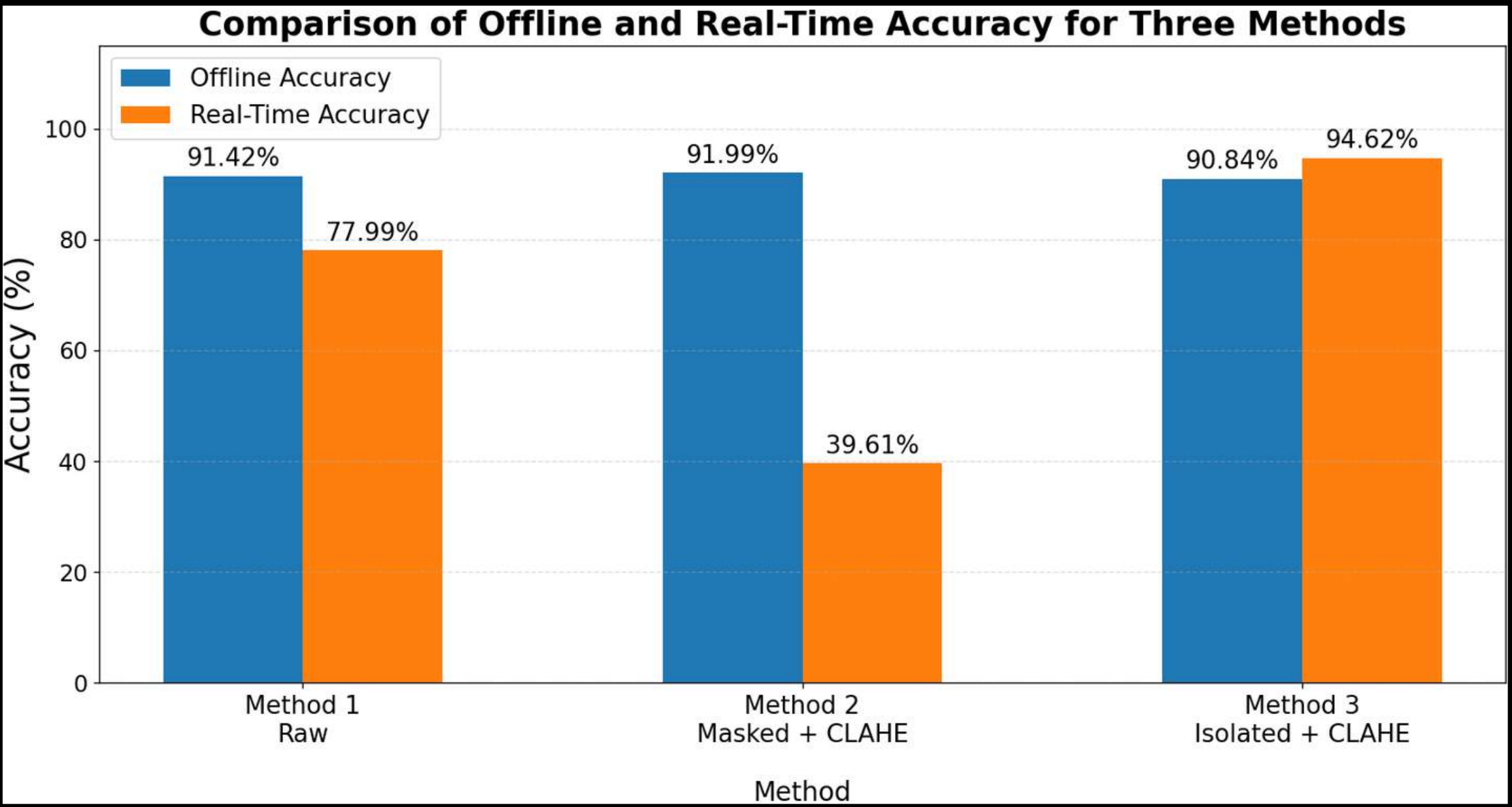
Method 1 worked reasonably well but was less stable than Method 3

➔ Moderate baseline performance

Method 1 worked reasonably well but was less stable than Method 3



Offline vs Real-Time Performance Comparison

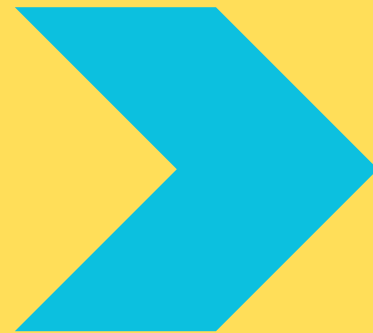


- ➔ **All three methods were similarly strong offline, with accuracy above 90%.**
- ➔ **Method 2 performed best offline, but dropped sharply in real-time testing.**
- ➔ **Method 3 performed best in real-time, showing stronger practical robustness.**

Discussion and Final Recommended Method

Offline

- Prepared test images
- Small performance gap
- Method 2 best overall metrics



Real Time

- Live camera + lighting change
- Larger performance gap
- Method 3 clearly best



Final Recommendation

- Method 3 Isolated + CLAHE
- Best balance of accuracy and stability



Conclusion

- Developed a YOLOv8 banana ripeness classification system for 4 classes
- Method 2 achieved the best offline performance
- Method 3 achieved the best real-time accuracy and stability
- Method 3 was selected as the final recommended method

Future Work

- Expand dataset with more images and lighting conditions
- Develop a mobile app for easier real-world use
- Improve preprocessing speed for real-time deployment
- Test in more complex supermarket environments



Thank You

Market Potential

Aldenaire & Partners

Global Banana Market Insights

Global banana market – \$7B+, growing demand for healthy snacks, with opportunities in retail, e-commerce, B2B, and exports.

Healthy Snacks, Expanding Opportunities

One of our banana plantations

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Aldenaire & Partners

Why Aldenaire & Partners

Comprehensive Banana Solutions

Banana expertise – farm-to-product solutions with customizable flavors, sustainable sourcing, and an experienced food and supply chain team.



Benefits for Customers & Partners



Convenience

Ready-to-eat and easy-to-use – convenient products designed for quick consumption and effortless preparation.



Health & Nutrition

Packed with natural energy, rich in fiber, and loaded with essential vitamins for a healthy and nutritious boost.



Brand Reliability

Reliable and consistent – ensuring high-quality products with great taste in every batch, every single time.



Partnership Opportunities

Flexible branding solutions – offering co-branding and private-label options to meet diverse business needs.